# Quentin Adolphe

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## EDUCATION

**Georgia Institute of Technology**
*Master of Science in Computer Science* | Edge AI specialization — Expected 2029

**Swarthmore College**
*Bachelor of Arts in Computer Science, Minor in Engineering* | GPA: 3.9/4.0 — Class of 2025

## PROFESSIONAL EXPERIENCE

**General Motors**
*Robotics AI/ML Software Engineer* — Sep 2026 - Present

- Designed the control-arbitration core of a ROS 2 cobot platform, granting exclusive robot ownership to one client at a time across REST, WebSocket, and gRPC, with latched HALT, confirmed-still recovery, and preemptive Stop/Cancel.
- Developing single-capture 6-DoF fixture localization, cross-checking ICP point-cloud fitting against fiducial tags.
- Built the cell's operator portal with a live URDF digital twin and hold-to-jog teleoperation with a dead-man stop.

**General Motors**
*Dimensional Systems Engineer* — Mar 2026 - Sep 2026

- Replaced a manual Excel/VBA workflow with a Databricks ETL pipeline, processing approximately 100,000 deliverable rows across vehicle programs to improve DSE visibility and execution.
- Built a 3D variation analysis tool for tolerance modeling, simulation, and reporting, saving 40+ hours per cycle.

**General Motors**
*CFD Automation Engineer* — Jul 2025 - Mar 2026

- Built a Python analytics framework to evaluate tire parameters affecting virtual vehicle stability.
- Rebuilt a legacy Tkinter GUI in React, Electron, and Flask, streamlining preprocessing, HPC jobs, and postprocessing.

**Swarthmore College**
*Teaching Assistant - Computer Vision & Mobile Robotics* — Jan 2024 - May 2025

- Helped students work through labs in perception algorithms, robot kinematics, and control systems.
- Reviewed and debugged assignments for 20+ students, giving feedback on code and problem-solving approaches.

**General Motors**
*Software Validation Engineer Intern* — May 2024 - Aug 2024

- Built applications supporting flash development, including a custom utility file generator.
- Automated ECU flash validation against Software-Defined Vehicle specifications.

**USAA**
*Software Engineering Intern* — May 2023 - Aug 2023

- Built React pages and database integration for a modernized full-stack life insurance application.

## PROJECTS

**Running Analysis with Machine Learning** | Kotlin, TensorFlow, Computer Vision, Machine Learning

- Used TensorFlow's Thunder convolutional neural network to estimate a sprinter's joint positions.
- Trained a neural network to identify the current phase of a sprinter in motion.

**3D Triangulation of Objects** | Python, Computer Vision, Stereo Calibration

- Calibrated a stereo camera setup to calculate 3D real-world coordinates from 2D images.

**Turtlebot Maze Traversal** | Python, Robotics, Dijkstra's Algorithm, Control Systems

- Implemented Dijkstra's algorithm and a proportional-derivative controller to navigate a robot through a maze.

## SKILLS

**Programming:** Python, Java, JavaScript, TypeScript
**Frameworks and Platforms:** React, Node.js, Flask, Databricks
**AI and Robotics:** Machine Learning, Computer Vision, ROS2, MCP Servers
**Data and Automation:** Data Aggregation, Process Automation, ETL Pipelines
**Certifications:** Six Sigma Green Belt